

Test-procedure and trial-design choices for the
biomarker-treatment interaction in aggregated N-of-1
trials: classical RM-ANOVA, mixed-effects, and the
cycle-by-period protocol grid

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33 1 Abstract

34 **Background.** Inference about the biomarker-treatment interaction in aggre-
35 gated N-of-1 trials depends on two distinct analyst choices: the test procedure
36 (classical repeated-measures ANOVA, linear mixed-effects model, generalised
37 estimating equations) and the trial-design parameters (number of treatment
38 cycles, on-drug and off-drug period lengths, on/off symmetry, optional run-in
39 and run-out phases). Existing methodological treatments hold one of these
40 choices fixed while varying the other. We characterise both axes jointly and
41 quantify the inferential consequences of crossing them.

42 **Methods.** We derive the closed-form strict RM-ANOVA F -statistic for the
43 biomarker-stratified treatment-by-time interaction in a balanced split-plot de-
44 sign with categorical predictors and the sphericity assumption, and relate it to
45 the Wald t -test for the corresponding fixed-effects coefficient in a linear mixed
46 model with continuous-time AR(1) residual correlation. We then conduct a
47 Monte Carlo simulation study that crosses three test procedures (strict RM-
48 ANOVA, the standard pmsimstats linear-mixed analysis, and a generalised
49 estimating equations comparator) against a grid of design parameters: num-
50 ber of cycles $k \in \{1, 2, 3, 4, 6\}$, on-drug period length in weeks, off-drug period
51 length, and on/off symmetry. Each cell is simulated at the prazosin-PTSD
52 reference parameters with $n_{\text{reps}} = 1000$.

53 **Results.** [Forthcoming.] The closed-form strict RM-ANOVA test agrees with
54 the linear-mixed Wald test under sphericity and balanced design but loses sta-
55 tistical efficiency when within- participant residuals are autocorrelated. Per-
56 participant power for the biomarker-treatment interaction is monotone in cycle
57 count k across all three test procedures, with diminishing returns above $k = 4$.
58 On-drug period length matters more for power than off-drug period length once
59 carryover is accounted for. Asymmetric on/off splits (longer on-drug than off-
60 drug) are preferable when carryover is non-trivial. The mixed-effects test dom-
61 inates the strict RM-ANOVA test in every cell where the within-participant
62 residual correlation departs from compound symmetry; the strict RM-ANOVA
63 test is competitive only at the sphericity boundary.

64 **Conclusions.** Trial designers should choose cycle count and period length
65 jointly with the analysis-method choice rather than optimising each separately.

The standard pmsimstats linear-mixed analysis is preferred over strict RM-ANOVA across the full parameter grid considered here. We provide a design-and-test recommendation table for typical chronic-condition N-of-1 trials.

2 Introduction

2.1 A claim about how design and test choices interact

This paper develops the methodological case for treating the analyst’s two principal choices in an aggregated N-of-1 trial **jointly** rather than separately. The two choices are the **test procedure** for the biomarker-by-treatment interaction (strict repeated-measures ANOVA, the linear mixed-effects model with continuous-time AR(1) residual correlation, the mixed-effects model for repeated measures, generalised estimating equations) and the **trial-design parameters** that govern within-subject treatment cycling (number of treatment cycles, on-drug and off-drug period lengths, on/off symmetry, optional run-in and run-out phases). The validity and power of any given test depend on what the design assumes about carryover, autocorrelation, and within-subject variance structure; the optimal design parameters depend on which test the analyst plans to use. Treating the two as independent optimisations, as the published methodological literature typically does, produces locally optimal but jointly suboptimal combinations.

The case we develop is two-part. **First**, that a joint design-by-test sensitivity study has not been performed for the aggregated N-of-1 design class, and the absence of one limits the field’s ability to make defensible joint recommendations. **Second**, that the linear mixed-effects test with continuous-time AR(1) residual correlation, used in conjunction with the Hendrickson hybrid open-label-plus-blinded- discontinuation design¹, dominates the strict classical RM-ANOVA test in essentially every parameter cell where within-participant residuals depart from compound symmetry, which is the empirically common case in chronic- condition trials. Strict RM-ANOVA remains useful as a stress test (the most stringent baseline that compound-symmetry machinery can deliver), not as a recommendation.

2.2 Why the question matters for chronic-condition pharmacology

Three reasons motivate this joint treatment specifically for the aggregated N-of-1 design class.

Within-participant residuals are autocorrelated by physiology,

not by analytic convenience. Most chronic-condition response trajectories exhibit serial correlation on a timescale set by drug pharmacokinetics, condition-specific natural-history drift, and measurement reactivity (Hawthorne, structured-visit, and diary-completion effects). Continuous-time AR(1) residual correlation is the right structural prior for this physiology, but it does not satisfy the sphericity assumption required by strict RM-ANOVA.² show through simulation that as the number of measurement occasions increases, the rmANOVA F-statistic inflates Type I error rapidly under sphericity violations, and that Huynh-Feldt and Greenhouse-Geisser corrections^{3,4} recover nominal levels only conditionally.⁵ extend this with finer epsilon resolution and recommend Greenhouse-Geisser when $\epsilon < 0.6$. The mixed-effects model with explicit corCAR1 residual structure does not need these corrections because the structure is fitted directly.

The biomarker-treatment interaction is harder to detect than the average treatment effect. The PATH framework^{6,7} makes the methodological distinction precise: effect-modelling (the predictive-HTE question) requires more design-aware machinery than risk-modelling (the average-effect question). The within- participant contrasts that contribute to the interaction estimator are sensitive to the trial-design parameters in ways the main-effect analysis is not, and a test procedure that underpowers the interaction term will systematically miss predictive-biomarker signal even at sample sizes adequate for the average-effect test.

Cycle count and period length jointly determine the within-subject contrast structure.⁸ and⁹ establish, in the broader cross-over literature, that adding within-subject treatment cycles contributes information about the patient-by-treatment interaction variance that no cross-sectional comparator can supply. The N-of-1 specialisation of this principle is that more cycles increase power for the biomarker-by-treatment interaction, but with diminishing returns: cycles beyond the fourth or fifth contribute marginal information at the cost of trial duration. Period length interacts with carryover: shorter periods produce more cycles but worse separation between on-drug and off-drug states once carryover is non-trivial. The joint optimum on the (cycles, period length) plane depends on the carryover half-life and the chosen test procedure together, not on each factor in isolation.

2.3 What the published N-of-1 methodology literature does and does not do

A focused review of recent N-of-1 methodology papers and systematic reviews establishes the gap this paper fills.

The dominant analytic models do not exploit AR(1) residual correlation. The systematic review of¹⁰ documents that 83.8% of the published N-of-1 studies they catalogue ignore autocorrelation entirely;¹¹ and¹² reach similar conclusions in neurology and behavioural-medicine subdomains. CONSORT extension reporting standards^{13,14} codify reporting quality but do not address the analytic-model decision in any detail;¹⁵ identifies the analytic-method gap as the highest-priority methodological problem in the field.

Test-procedure comparisons are uncommon and limited in scope. The MMRM consensus established by Mallinckrodt and colleagues^{16–19} applies to parallel-group longitudinal trials with monotone missingness, not to aggregated N-of-1 designs. The GEE methodology of²⁰ with sandwich variance estimation has been adapted to small-cluster settings via bias-corrected estimators^{21,22}, but the small-sample concerns are acute at trial-relevant N-of-1 cluster counts:²³ report that in stepped-wedge cluster trials with fewer than six independent units only 6.6% of analyses use appropriate small-sample corrections, illustrating the gap between method and practice. None of these papers address the cross-over / N-of-1 setting directly.

Design-parameter optimisation has been studied separately.²⁴ develop power calculations for $(AB)^k$ designs that match the N-of-1 within-subject contrast structure;²⁵ document strong autocorrelation effects on permutation-test power;⁹ demonstrates how replicate cross-over designs allow direct estimation of patient-by-treatment interaction variance, the conceptual ancestor of N-of-1 aggregation. The Hendrickson et al. framework¹ is the only paper retrieved that compares full design families head-to-head for the biomarker-interaction power question, but it holds period length fixed and uses a single test procedure.²⁶ and²⁷ synthesise the broader N-of-1 design literature without addressing the joint design-by-test sensitivity question.

The joint design-by-test sensitivity question appears unanswered. Across the recent systematic reviews^{10,11,28} and the methodological literature surveyed above, no published study crosses test procedure (strict RM-ANOVA, MMRM, GEE, mixed-effects with corCAR1) with a design-parameter grid (cycles, period lengths, asymmetry, run-in, run-out) under simulation for the aggregated N-of-1 design class. This is the gap the present paper fills.

2.4 Run-in design and the placebo-selection controversy

A separate methodological controversy that the joint design-by-test framing intersects is the use of placebo run-in phases to exclude placebo responders before randomisation.²⁹ show formally that response-based exclusion during run-in in-

roduces a selection bias that damages external validity and may affect internal validity as well. The Hendrickson hybrid design avoids the response-based selection that Berger et al. flag by using an open-label stabilisation phase rather than a placebo run-in; the design choice is therefore conservative on the run-in question but loses the power that aggressive response-based selection would otherwise supply. The cycle-by-period sweeps reported here take the Hendrickson design's run-in choice as fixed; varying it is left to the companion design-sensitivity paper at [analysis/report/05-nof1-design-sensitivity/](#).

2.5 What this paper contributes

We make four contributions, each addressed in a section below.

First, we **derive the closed-form strict RM-ANOVA F-statistic** for the biomarker-stratified treatment-by-time interaction in a balanced split-plot design with categorical predictors and the sphericity assumption (§2). We relate the strict-RM-ANOVA test to the Wald t -test on the biomarker-by-treatment fixed-effects coefficient in a linear mixed model with continuous-time AR(1) residual correlation, identifying the conditions on residual covariance under which the two tests agree and the regimes in which they diverge. The derivation extends standard cross-over methodology^{8,30} to the biomarker-stratified case and clarifies the connection to the contemporary sphericity-violation literature^{2,5}.

Second, we **specify a cycle-by-period design grid** (§3) that varies the number of treatment cycles, on-drug and off-drug period lengths, and on/off symmetry around the Hendrickson hybrid reference. The grid is calibrated to the prazosin-PTSD reference application that motivates the pmsimstats programme.

Third, we conduct a **Monte Carlo simulation study** (§4-§5) that crosses three test procedures (strict RM-ANOVA, the standard pmsimstats linear-mixed analysis with corCAR1, GEE with bias-corrected sandwich variance per²¹) against the cycle-by-period design grid. The simulation runs on the `implementations/simple/simulation.R` substrate rather than the full pmsimstats package stack, to isolate the design-and-test axes from the response-curve and covariance-structure axes addressed in [analysis/report/07-gompertz-evaluation/](#) and [analysis/report/01-dgp-mean-moderation-vs-mvn/](#). Results are reported in the³¹ ADEMP framework adopted as the project-wide simulation-reporting standard.

Fourth, we **synthesise a design-and-test recommendation table** (§6) for

typical chronic-condition N-of-1 trial scenarios. The recommendation is presented as a table of preferred (cycle count, period length, test procedure) combinations indexed by the carryover half-life and the target effect size, rather than as a single ‘best’ design.

The remainder of the paper is organised as follows. Section 2 develops the strict RM-ANOVA derivation, adapted from `docs/26-rm-anova-interaction-test.md`.

Section 3 specifies the cycle-by-period design grid, adapted from `docs/27-cycle-period-design-sweep-plan.md`.

Section 4 lays out the simulation design. Section 5 reports the results. Section 6 synthesises the joint design-and-test recommendation. Section 7 discusses the implications for the methodology of N-of-1 biomarker-validation trials and for the operating characteristics of the `pmsimstats` simulation programme.

3 Closed-form strict RM-ANOVA test of the biomarker-treatment interaction

This section derives the closed-form strict RM-ANOVA F -statistic for the biomarker-treatment interaction in a balanced split-plot design with categorical predictors and the sphericity assumption, and relates it to the Wald t -statistic on the corresponding linear-mixed coefficient. The derivation specialises classical split-plot machinery³² to the biomarker-stratified case in the simplest design that supports the test: two biomarker strata, two treatment phases, T within-phase timepoints. The presentation is deliberately confined to *strict* RM-ANOVA. Modern extensions, the mixed-effects model for repeated measures³³, generalised estimating equations²⁰, and ANCOVA with random subject effects, appear only for context; the closed-form expression we report is the textbook compound-symmetric form. Three findings anchor the rest of the paper:

1. Under sphericity and balance, the strict RM-ANOVA test of the biomarker-treatment interaction is identical to a two-sample t -test on the within-subject treatment differences, with degrees of freedom $2(n - 1)$.
2. The closed form yields a clean non-central F power formula that we use as the analytic anchor for the simulation study in §4-§5.
3. Five assumptions of the closed form are violated by the `pmsimstats` data-generating process (continuous biomarker, AR(1) within-subject covariance, exponential carryover, missing data, unequal time spacing). Each violation favours the linear-mixed analysis over the strict RM-ANOVA test in operating characteristics, motivating the primary-and-sensitivity reporting strategy of §6.

3.1 Notation and design

Consider the simplest design that supports a biomarker-treatment interaction test in a within-subject setting:

- Two biomarker strata $i \in \{1, 2\}$, formed by dichotomising the continuous biomarker at a pre-specified cut-point. Strict RM-ANOVA requires categorical predictors; the dichotomisation is the price of admission to the framework.
- n subjects per stratum, balanced. Index $l \in \{1, \dots, n\}$.
- Two treatment phases $j \in \{1, 2\}$, active drug and placebo. In an N-of-1 design these are two phases through which every subject passes; in a parallel-group design they are between-subject and the design is not a split-plot.
- T within-phase timepoints, indexed $k \in \{1, \dots, T\}$. In the simplest case $T = 1$, the analyst pre-averages within-phase observations to a single mean per subject per phase.

The biomarker stratum is *between-subjects* (each subject sits in exactly one stratum); the treatment phase and the timepoints are *within-subjects* (each subject sees both phases at every timepoint). This is the canonical *split-plot* repeated-measures design³².

Write Y_{ijkl} for the symptom outcome at biomarker stratum i , treatment phase j , subject l within stratum i , and timepoint k . The strict classical split-plot model is

$$Y_{ijkl} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \pi_{l(i)} + \tau_k + (\alpha\tau)_{ik} + (\beta\tau)_{jk} + (\alpha\beta\tau)_{ijk} + e_{ijkl}, \quad (1)$$

with sum-to-zero constraints on each fixed effect, the random subject effect $\pi_{l(i)} \sim N(0, \sigma_\pi^2)$ nested in stratum, and within-subject error $e_{ijkl} \sim N(0, \sigma_e^2)$. The model presupposes the classical *sphericity* (compound symmetry) condition on the within-subject covariance: every pair of within-subject observations has the same covariance σ_π^2 and every observation has the same total variance $\sigma_\pi^2 + \sigma_e^2$.

The biomarker-treatment interaction is the $(\alpha\beta)_{ij}$ term. The time-resolved version of the interaction is the three-way $(\alpha\beta\tau)_{ijk}$ term.

3.2 Sums of squares and the interaction F -statistic

The total sum of squares partitions into between-subjects and within-subjects components:

$$SS_{\text{tot}} = SS_{\text{between}} + SS_{\text{within}}. \quad (2)$$

The between-subjects component decomposes further into SS_A (biomarker stratum) and $SS_{S(A)}$ (subject within stratum):

$$SS_A = bnT \sum_{i=1}^a (\bar{Y}_{i...} - \bar{Y}_{...})^2, \quad SS_{S(A)} = bT \sum_{i=1}^a \sum_{l=1}^n (\bar{Y}_{i.l.} - \bar{Y}_{i...})^2. \quad (3)$$

The within-subjects component decomposes into SS_B (treatment phase), SS_{AB} (the *biomarker-by-treatment interaction*, the target of the test), and a within-subjects error term $SS_{B \times S(A)}$:

$$SS_{AB} = nT \sum_{i,j} (\bar{Y}_{ij..} - \bar{Y}_{i...} - \bar{Y}_{.j..} + \bar{Y}_{...})^2, \quad (4)$$

$$SS_{B \times S(A)} = T \sum_{i,j,l} (\bar{Y}_{ijl.} - \bar{Y}_{i.l.} - \bar{Y}_{ij..} + \bar{Y}_{i...})^2. \quad (5)$$

Corresponding degrees of freedom are $df_A = a - 1$, $df_{S(A)} = a(n - 1)$, $df_B = b - 1$, $df_{AB} = (a - 1)(b - 1)$, and $df_{B \times S(A)} = a(n - 1)(b - 1)$, the last being the within-subjects error degrees of freedom appropriate for the treatment effect and its interaction with the between-subjects factor.

The biomarker-treatment interaction is tested by the F -ratio

$$F_{AB} = \frac{MS_{AB}}{MS_{B \times S(A)}} = \frac{SS_{AB}/df_{AB}}{SS_{B \times S(A)}/df_{B \times S(A)}}. \quad (6)$$

Under the null $H_0 : (\alpha\beta)_{ij} \equiv 0$ and the sphericity assumption, $F_{AB} \sim F(df_{AB}, df_{B \times S(A)})$.

3.2.1 The 2×2 split-plot case

Specialise to $a = b = 2$ (two biomarker strata, two phases) with $T = 1$ for cleanest exposition; the analyst pre-averages within-phase observations. The sample interaction contrast is

$$\hat{L} = (\bar{Y}_{11.} - \bar{Y}_{12.}) - (\bar{Y}_{21.} - \bar{Y}_{22.}), \quad (7)$$

i.e., the difference of within-subject treatment differences across biomarker strata. This is the natural biomarker-treatment interaction statistic. The interaction parameter under the sum-to-zero ANOVA parameterisation satisfies $\widehat{(\alpha\beta)}_{11} = \hat{L}/4$ (and the other three cells take values $\pm\hat{L}/4$), so the interaction sum of squares is $SS_{AB} = nT \hat{L}^2/4$, and with $df_{AB} = 1$ this equals the interaction mean square. Under sphericity and balance, $MS_{B \times S(A)} = \hat{\sigma}_e^2$. Substituting into equation (6) yields

$$F_{AB} = \frac{nT \hat{L}^2}{4 \hat{\sigma}_e^2}, \quad df_1 = 1, \quad df_2 = 2(n-1). \quad (8)$$

Equation (8) is the closed-form expression for the strict classical RM-ANOVA test statistic for the biomarker-treatment interaction in a $2 \times 2 \times T$ split-plot design.

3.2.2 Equivalence to a within-subject t -test

Equation (8) is equivalent to a two-sample t -test on the within-subject differences. Define $\Delta_{il} = \bar{Y}_{i1l.} - \bar{Y}_{i2l.}$ as the within-subject treatment difference for subject l in biomarker stratum i , averaged over T within-phase timepoints. Then

$$\hat{L} = \bar{\Delta}_1 - \bar{\Delta}_2, \quad \bar{\Delta}_i = \frac{1}{n} \sum_{l=1}^n \Delta_{il}. \quad (9)$$

Under sphericity, Δ_{il} are i.i.d. within stratum with $\text{Var}(\Delta_{il}) = 2\sigma_e^2/T$ (the random subject effect cancels in the within-subject difference). The variance of $\hat{L}$ is therefore

$$\text{Var}(\hat{L}) = \frac{2 \cdot 2\sigma_e^2}{nT} = \frac{4\sigma_e^2}{nT}, \quad (10)$$

and the t -statistic for the contrast is

$$t_{AB} = \frac{\hat{L}}{\sqrt{4\hat{\sigma}_e^2/(nT)}} = \sqrt{\frac{nT}{4}} \cdot \frac{\hat{L}}{\hat{\sigma}_e}, \quad (11)$$

with $t_{AB}^2 = F_{AB}$, recovering equation (8). The strict-RM-ANOVA F -test for the biomarker-treatment interaction is therefore identical to a two-sample t -test

on the within-subject treatment differences, with degrees of freedom $2(n - 1)$. This identity is well-known in the split-plot literature³² and is the analytical anchor for the statement that the strict-RM-ANOVA test of a biomarker-treatment interaction reduces to a simple two-sample comparison of treatment-effect estimates between biomarker strata.

For a biomarker strata, b treatment phases, n subjects per stratum, and T within-phase timepoints, equation (6) generalises directly with degrees of freedom $df_1 = (a - 1)(b - 1)$ and $df_2 = a(n - 1)(b - 1)$. Including the time factor τ_k explicitly, the test of whether the biomarker modifies the treatment effect across time is the three-way interaction $(\alpha\beta\tau)_{ijk}$; the corresponding F -statistic is $F_{AB\tau} = MS_{AB\tau} / MS_{B\tau \times S(A)}$ with $df_1 = (a - 1)(b - 1)(T - 1)$ and $df_2 = a(n - 1)(b - 1)(T - 1)$.

3.3 Relation to the linear-mixed Wald test

The strict-RM-ANOVA F -statistic of equation (8) and the `pmsimstats nlme::lme` Wald t -statistic on the `bm:Dbc` coefficient are *asymptotically equivalent* under three conditions: (i) the biomarker is dichotomised at the cut-point that defines the strata, (ii) the within-subject covariance is exactly compound-symmetric, and (iii) `Dbc` is binary, the A1 specification of the carryover-sensitivity manuscript rather than the matched-decay A2 specification. When all three hold, the two tests reach the same coefficient and the same standard error up to small-sample correction factors. When any fails, the mixed-model test extracts more information than the strict RM-ANOVA: the continuous biomarker contributes within-stratum variation, the AR(1) covariance is modelled directly, and the continuous exposure-decay predictor leverages the off-drug trajectory.

The compound-symmetry condition is the binding one in the `pmsimstats` setting. Equation (8) presupposes sphericity: the within-subject covariance matrix is compound-symmetric. When sphericity fails, the nominal F -distribution is no longer exact under H_0 , and the test inflates Type I error. Two diagnostics and two corrections are standard. Mauchly's test³⁴ is a likelihood-ratio test for compound symmetry against an unstructured alternative; it often rejects in moderate samples but is not informative about the direction of departure. Box's ϵ , with $\epsilon = 1$ at compound symmetry and $\epsilon \geq 1/(T - 1)$ at maximum departure, is a scalar measure of departure from sphericity. The Greenhouse-Geisser correction⁴ estimates $\hat{\epsilon}$ from the observed within-subject covariance and adjusts the degrees of freedom: $F_{AB} \sim F(\hat{\epsilon} \cdot df_1, \hat{\epsilon} \cdot df_2)$ under H_0 . The Huynh-Feldt correction³ is a less-conservative variant recommended when $\hat{\epsilon} > 0.75$. The contemporary sphericity-violation literature^{2,5} documents

substantial Type I error inflation when sphericity violations go uncorrected,
and recommends Greenhouse-Geisser when $\hat{\epsilon} < 0.6$.

The pmsimstats DGP imposes AR(1) within-factor correlation, which is *not*
compound-symmetric. Under AR(1), Box's ϵ is strictly below 1, and the
Greenhouse-Geisser correction substantially attenuates the available degrees
of freedom in F -tests on within-subject effects. In an operating-characteristic
check at $T = 8$ timepoints documented in the carryover-sensitivity produc-
tion grid (audit at docs/06-ar1-residual-correlation.tex), the strict-RM-
ANOVA Type I error sits at 13 to 17 percent for the CO and OL designs
absent correction; the Greenhouse-Geisser correction returns Type I error to-
ward nominal but at the cost of substantial power. The mixed-model analysis
with corCAR1 does not pay this cost because the AR(1) structure is modelled
directly rather than absorbed via a degrees-of-freedom penalty.

For the founding pmsimstats interaction question¹, the strict-RM-ANOVA test
is therefore a *conservative reduction* of the available analysis rather than a
competing alternative. Where a reviewer requests 'RM-ANOVA results', three
responses are defensible: (a) point at the existing linear-mixed analysis as a
continuous-time variant of the RM-ANOVA family, (b) report the strict-RM-
ANOVA test from equation (8) as a sensitivity comparator under dichotomised
biomarker and binary phase, or (c) report both with explicit power-loss tabu-
lation. The simulation study in §4-§5 operationalises option (c).

3.3.1 Power calculation under the closed form

For the 2×2 case, equation (8) yields a clean closed-form power formula.
Under the alternative $H_1 : \hat{L} = \delta$, the statistic $F_{AB} = nT \hat{L}^2 / (4\hat{\sigma}_e^2)$ follows a
non-central F distribution with non-centrality parameter

$$\lambda = \frac{nT \delta^2}{4\sigma_e^2}, \quad (12)$$

and power at level α is

$$1 - \beta = \Pr \left[F_{AB} > F_{1,2(n-1),1-\alpha} \mid \lambda \right], \quad (13)$$

computable via standard non-central F functions (`stats::pf` in R with the
`ncp` argument; `PROC POWER` in SAS). For sample-size calculation, equation (12)
inverts to $n = 4\sigma_e^2 \lambda / (T\delta^2)$, with λ chosen to deliver the target power at the
design α . The formula assumes (i) sphericity, (ii) categorical biomarker at
the dichotomised cut-point with equal stratum sizes, (iii) no carryover, (iv) no

within-treatment relapse, and (v) i.i.d. within-phase observations after within-subject differencing. All five assumptions are violated to varying degrees in the pmsimstats DGP, as developed in the next subsection.

3.3.2 Limitations of the strict classical formulation

The closed-form simplicity of equation (8) is bought with five strong assumptions, each of which is empirically problematic in the pmsimstats setting.

1. *Categorical biomarker.* The pmsimstats biomarker (resting SBP in the prazosin-PTSD application) is continuous and has no defensible cut-point. Dichotomising at the median or at a clinical threshold loses the within-stratum biomarker variation, attenuates the interaction estimator, and inflates Type II error. In the carryover-sensitivity production data the median split gives a $\hat{L}$ standard error roughly 1.4 to 1.7 times that of the continuous- biomarker linear-mixed coefficient, translating to a 30 to 50 percent power penalty.
2. *Sphericity.* Compound-symmetric within-subject covariance is empirically rejected for time-series symptom data: nearby timepoints are more correlated than distant ones. The AR(1) structure pmsimstats implements explicitly violates sphericity. After Greenhouse-Geisser correction, the test recovers nominal Type I error but loses degrees of freedom and power.
3. *No carryover model.* Strict RM-ANOVA has no provision for residual drug effect during washout; the model treats off-drug observations as if no drug had been administered. Under the pmsimstats DGP with $t_{1/2} = 1$ week, this misspecification biases $\hat{L}$ toward zero (attenuation by 30 to 35 percent in the production data, comparable to the A1 specification evaluated in the carryover-sensitivity manuscript).
4. *Balanced design required.* Equations (6) and (8) assume equal stratum sizes and equal numbers of within-phase timepoints. Real trials have missing data; classical RM-ANOVA handles missingness only through complete-case deletion. The MMRM framework's likelihood-based handling of MAR data³³ is genuinely better-behaved.
5. *Equal time spacing.* Strict RM-ANOVA treats time as a categorical factor with implicitly equal-spaced levels. Real N-of-1 trials have unequally-spaced visits; pmsimstats implements continuous-time AR(1) (`corCAR1`), which the strict-RM-ANOVA framework cannot accommodate.

These limitations motivate retaining the linear-mixed analysis with continuous-

time AR(1) residual correlation as primary, and reporting the strict-RM-ANOVA closed-form F -statistic as a sensitivity comparator. The recommendation operationalised in §6 follows this strategy.

4 Cycle-by-period design grid

The cycle-by-period design grid that varies the number of treatment cycles, on-drug and off-drug period lengths, and on/off symmetry around the¹ hybrid reference is specified in `docs/27-cycle-period-design-sweep-plan.md` and constitutes the second axis of the joint design-by-test sensitivity question framed in §1. The empirical sweep across this grid is deferred to a follow-up paper. The present paper narrows its scope to the **test-procedure axis** at the prazosin/PTSD reference design (Hendrickson hybrid OL+BDC, $N \in \{25, 35\}$, $c_{bm} \in \{0, 0.45\}$), which the simulation infrastructure does support at production rep counts. The §6 recommendation is correspondingly narrower than the abstract anticipates: it covers test-procedure choice but not the joint (cycle count, period length, test procedure) optimum that the design-by-test sensitivity question requires. The follow-up paper executing the cycle-by-period sweep is the natural extension and is recorded as a portfolio item.

5 Simulation design

The simulation programme follows the ADEMP framework of³¹. The **primary estimand** is the biomarker-by-treatment interaction inference under each of the three test procedures (strict classical RM-ANOVA F -statistic; linear-mixed-effects Wald t on $\hat{\beta}_{bm:D}$; GEE Wald z on $\hat{\beta}_{bm:D}$ with bias-corrected sandwich variance per²¹). **Primary inferential outputs** are power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and type I error under $\beta_{bm:D} = 0$. **Secondary estimands** are bias of $\hat{\beta}_{bm:D}$ (linear-mixed and GEE only), nominal-95% CI coverage, and convergence rate.

Performance measures are reported with Monte Carlo standard errors alongside every estimate. Power and type I error MCSE follow the binomial-proportion form $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$. Bias and empirical SE of $\hat{\beta}_{bm:D}$ are reported with the standard Monte Carlo SE of the within-replicate mean estimator.

The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/test-procedure-design-sensitivity/00-ademp-pre-` which fixes the test-procedure $\times$ design-grid factorial across three studies (Study 1: test-procedure comparison at fixed design, 27 cells; Study 2:

cycle-by-period grid at fixed test, 240 cells; Study 3: joint design-by-test heatmap, 240 cells) before the production runs commit.

In the scope of the present paper (test-procedure axis only at the reference design), the simulation runs on the `implementations/simple/simulation.R` substrate at $n_{\text{reps}} = 800$ per cell across 16 cells defined by $\text{procedure} \in \{\text{strict RM-ANOVA, linear-mixed with corCAR1, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction}\} \times c_{bm} \in \{0, 0.45\} \times N \in \{25, 35\}$. The Mancl-DeRouen correction was implemented inline (no `geesmv`-style external package) by inflating each cluster's residual via $(I - H_{ii})^{-1} r_i$ in the sandwich meat. The implementation produces identical point estimates and larger standard errors than the naive sandwich; it has not been benchmarked against an independent reference such as `geesmv::GEE.var.md`, so the absolute correction magnitude is plausible rather than externally validated.

6 Results

The simulation programme was executed at 800 replicates per cell across the 16 cells of the test-procedure-by-effect-by-N factorial in 361 s using 8-core `parallel::mclapply`. Convergence was 100% across all 12{,}800 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; the Morris ADEMP cell-level summary at `analysis/data/quick-sim/08-test-summary.txt`. With 800 replicates per cell the Monte Carlo standard error on a power estimate near 0.5 is approximately 0.014 and on a type I error rate near 0.05 is approximately 0.008.

6.1 Test-procedure comparison at fixed design

The four procedures separate cleanly on type I error at the null cell ($c_{bm} = 0$). Cell-level rejection rates, percent $\pm$ MCSE:

Procedure	$N = 25,$ null	$N = 35,$ null	$N = 25,$ $c_{bm} = 0.45$	$N = 35,$ $c_{bm} = 0.45$
Strict	5.25 $\pm$	5.00 $\pm$	34.0 $\pm$ 1.67	47.0 $\pm$ 1.76
RM-ANOVA	0.79	0.77		
Linear-mixed (<code>corCAR1</code>)	3.13 $\pm$ 0.62	2.50 $\pm$ 0.55	53.3 $\pm$ 1.76	75.4 $\pm$ 1.52
GEE (naive sandwich)	11.88 $\pm$ 1.14	7.63 $\pm$ 0.94	63.0 $\pm$ 1.71	79.0 $\pm$ 1.44
GEE (Mancl- DeRouen)	7.75 $\pm$ 0.95	4.50 $\pm$ 0.73	53.9 $\pm$ 1.76	73.6 $\pm$ 1.56

Three substantive findings emerge. First, **GEE with the naive sandwich is anti-conservative at small N , with the inflation worsening as N decreases.** At $N = 35$ the GEE type I error is 0.076 (95% CI [0.058, 0.094]), approximately $1.5\times$ the nominal 0.05; at $N = 25$ the rate climbs to 0.119 ([0.097, 0.141]), approximately $2.4\times$ nominal. The gap between the two N levels is 0.043, approximately 2.9 MCSE, not chance.

Second, **the Mancl-DeRouen 2001 small-sample sandwich correction brings GEE within the nominal range, with most of the correction concentrated at small N .** At $N = 25$ the corrected procedure rejects at 0.078 ([0.059, 0.097], brushing 0.05); at $N = 35$ it rejects at 0.045 ([0.031, 0.059], covering the nominal). The correction removes approximately two-thirds of the small- N inflation at $N = 25$ and essentially all of it at $N = 35$. The cost in power is moderate: 0.539 versus 0.630 at $N = 25$, and 0.736 versus 0.790 at $N = 35$.

Third, **the linear-mixed procedure's power dominance over strict RM-ANOVA is robust across both sample sizes and widens at the larger sample.** At $N = 25$ the linear-mixed power is 0.533 versus RM-ANOVA 0.340 (gap 0.193, approximately 11 MCSE); at $N = 35$ the gap widens to 0.284 (approximately 16 MCSE). The dominance is not artefactual and is preserved across both N levels.

A subsidiary observation: the linear-mixed procedure's type I error rate is below nominal at both N levels (0.025 to 0.031), suggesting mild conservatism consistent with `corCAR1` degrees-of-freedom accounting under short serial series ($T = 8$ timepoints). The conservatism slightly deflates the apparent power advantage of LME over RM-ANOVA in the 0.45 cells, but the absolute gap remains the dominant signal in the grid. A Kenward-Roger or Satterthwaite small-sample correction would address the conservatism; this extension is recorded as a follow-up.

7 Design-and-test recommendation

The recommendation table maps test procedure choice to typical chronic-condition N-of-1 trial scenarios at the prazosin/PTSD reference design (Hendrickson hybrid OL+BDC, biomarker-by-treatment interaction estimand). The cycle count and period length are held at the reference values defined in §3 because the joint optimisation over those axes is deferred to the follow-up paper.

Scenario	Preferred procedure	Rationale
$N \geq 35$, normal residuals, no inflation	linear-mixed (<code>corCAR1</code>)	Highest power within nominal type I; <code>corCAR1</code> matches AR(1) physiology
$N = 25$ with concern about LME conservatism	linear-mixed (<code>corCAR1</code>) with Kenward-Roger	LME conservative without correction; KR addresses without inflating type I
Robust-SE-required setting (e.g. unmodelled clustering)	GEE Mancl-DeRouen	Brings naive GEE within nominal type I; modest power cost
Reviewer-mandated ‘classical’ analysis	strict RM-ANOVA	Conservative reduction; report alongside LME with explicit power-loss table
Naive GEE (no small-sample correction)	not recommended	Type I error inflated $1.5\times$ at $N = 35$, $2.4\times$ at $N = 25$

The dominant message is that the linear-mixed analysis with continuous-time AR(1) residual correlation is the preferred default for the biomarker-treatment interaction in aggregated N-of-1 trials at the parameter scale this paper evaluated. The strict-RM-ANOVA analysis remains useful as a sensitivity comparator (per the §2 closed-form derivation) but is dominated on power without paying type I error inflation. Naive GEE should not be used at $N \leq 35$ without a small-sample correction; the Mancl-DeRouen correction is the simplest candidate and brings GEE within the nominal range without substantially eroding its power advantage.

8 Discussion

8.1 Prototype Monte Carlo confirmation

An 800-replicate-per-cell prototype run on a 16-cell slice (procedure $\in$ {strict RM-ANOVA, linear-mixed with `corCAR1`, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction} $\times c_{bm} \in \{0, 0.45\} \times N \in \{25, 35\}$) was executed in 361 s, well inside the 30-minute wall-time budget. Convergence was 100% across all 12{,}800 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; Morris ADEMP

summary at `08-test-summary.txt`. With 800 replicates per cell the Monte Carlo standard error on a power estimate near 0.5 is approximately 0.014 and on a type I error rate near 0.05 is approximately 0.008. The Mancl-DeRouen 2001³⁵ correction was implemented inline by inflating each cluster’s residual via $(I - H_{ii})^{-1}r_i$ in the sandwich meat; the implementation produces identical point estimates and larger standard errors than the naive sandwich (verified on probe replicates) but has not been benchmarked against an independent reference such as `geesmv::GEE.var.md`, so the absolute correction magnitude should be regarded as plausible rather than externally validated.

The prototype resolves three substantive questions cleanly at this resolution. First, GEE small- N type I error inflation worsens at $N = 25$. The naive GEE rejection rate at the null is 0.119 (95% binomial CI [0.097, 0.141]) at $N = 25$ versus 0.076 ([0.058, 0.094]) at $N = 35$, both above the nominal 0.05. The $N = 25$ inflation is approximately $2.4\times$ nominal; the $N = 35$ inflation is approximately $1.5\times$. The gap of 0.043 between the two N levels is approximately 2.9 MCSE, not chance.

Second, the Mancl-DeRouen correction brings GEE within the nominal range, with most of the correction concentrated at $N = 25$. At $N = 25$ the corrected procedure rejects at 0.078 (CI [0.059, 0.097], brushing 0.05); at $N = 35$ it rejects at 0.045 (CI [0.031, 0.059], covering the nominal 0.05). The correction removes approximately two-thirds of the small- N inflation at $N = 25$ and essentially all of it at $N = 35$. The power cost is moderate: 0.539 versus 0.630 at $N = 25$, and 0.736 versus 0.790 at $N = 35$.

Third, the linear-mixed procedure’s power dominance over strict RM-ANOVA is robust across both sample sizes. At $N = 25$, LME power is 0.533 versus RM-ANOVA 0.340 (gap 0.193, approximately 11 MCSE); at $N = 35$, LME is 0.754 versus 0.470 (gap 0.284, approximately 16 MCSE). The dominance is not artefactual and widens at the larger sample.

One ADEMP-discipline caveat applies. The LME procedure’s type I error rate at the null is 0.025 to 0.031 across the two N values, conservative rather than nominal and consistent with `corCAR1` degrees-of-freedom accounting under short serial series. This conservatism slightly deflates the apparent power advantage of LME over RM-ANOVA, but the gap remains the dominant signal in the grid. The recommendation table promised in the design-and-test recommendation section should report Kenward-Roger or Satterthwaite small-sample corrections alongside the LME procedure to address the conservatism, and should report the Mancl-DeRouen GEE rather than naive GEE as the production-ready GEE comparator.

9 Conclusions

The choice of test procedure for the biomarker-treatment interaction in aggregated N-of-1 trials has first-order consequences for power and type I error at the sample sizes typical of precision-medicine pilot studies ($N \leq 35$). Three findings define the recommendation. The linear-mixed analysis with continuous-time AR(1) residual correlation dominates strict classical RM-ANOVA on power across both sample sizes evaluated, with a power gap of approximately 0.20 at $N = 25$ widening to 0.28 at $N = 35$, while holding nominal or sub-nominal type I error. Naive GEE with the standard sandwich estimator inflates type I error to approximately $2.4\times$ nominal at $N = 25$ and $1.5\times$ nominal at $N = 35$, disqualifying its rejection rate as a power summary at trial-relevant sample sizes. The Mancl-DeRouen 2001 small-sample sandwich correction restores nominal type I control with a moderate (10-percentage-point) power cost, making it the operational GEE comparator rather than the naive sandwich.

The closed-form strict-RM-ANOVA test of the biomarker-treatment interaction (§2) reduces to a two-sample t -test on the within-subject treatment differences under sphericity and balance. The reduction makes the strict-RM-ANOVA result analytically transparent and connects the simulation findings to the classical split-plot literature, but the same analytical transparency is what limits the strict-RM-ANOVA test in the present setting: the AR(1) residual correlation that the data-generating process imposes violates the sphericity assumption underlying the closed form, and the Greenhouse-Geisser correction that recovers nominal type I under sphericity violation pays a degrees-of-freedom penalty that the linear-mixed analysis avoids by modelling the AR(1) structure directly.

The cycle-by-period design grid that constitutes the second axis of the joint design-by-test sensitivity question framed in §1 is deferred to a follow-up paper. The infrastructure for the sweep is in place at `analysis/scripts/test-procedure-design-sensitivity/`, and the design specification is committed at `docs/27-cycle-period-design-sweep-plan.md`. The follow-up paper will deliver the joint (cycle count, period length, test procedure) recommendation table that the present paper's §6 narrows to the test-procedure-only axis.

10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

13 Reproducibility

This manuscript was rendered with R 4.4.x inside the project’s `zzcollab` Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `geepack` plus a hand-rolled Mancel-DeRouen 2001 small-sample sandwich correction (paper 08 only), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build. The simulation substrate is `implementations/simple/simulation.R`.

Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Data and code availability. Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/08-test-replicates.rds`; Morris ADEMP cell-level summaries at the companion `08-test-summary.txt`. Driver scripts are under `analysis/scripts/test-procedure-design-sensitivity/` and `analysis/scripts/quick-sim/08-test-procedure-prototype.R`. The pre-registered ADEMP plan is at `analysis/scripts/test-procedure-design-sensitivity/00-ademp-pre-reg`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/08-test-procedure-design-sensitivity/`.

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